

Can Structured Review Help a Market of Language Model Traders?

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

This account connects a study of language model traders in a double auction with a study of reviewer and critic agents for code review. The thread asked whether reviewing a trader's proposed quote could improve the market's total gains from trade. The peers found that a better price for one trader need not raise those gains, and that even an extra profitable trade can lower them under the auction's stated values. The thread ended as a draft because these checks sharpen the question but neither paper tests review in the auction.

1 Introduction

Struski and colleagues study language model agents buying and selling a single good in a continuous double auction [6]. Buyers submit bids, sellers submit asks, and a trade occurs when an offer meets a standing quote. Each agent knows its own limit for a profitable trade, while the full pattern of buyer values and seller costs is hidden from it. The paper compares the resulting prices and total gains from trade with a human benchmark. It reports slower or absent convergence to the competitive price and less efficient allocations than in the human market.

Qiu and Gill study a small team for code review [5]. A main agent produces an artefact, a reviewer comments on it, and a critic challenges that review before the main agent edits. Their first protocol can produce agreement without enough evidence. On a code review benchmark, a revised prompt that requires the critic to distinguish evidence from concern improves review quality. The paper does not test that revised prompt in its repository repair experiment.

The proposed connection was to freeze a trader's intended quote while a reviewer and critic examine it, then let the trader decide what to submit. The market paper gives a concrete setting: some traders make tiny quote improvements until a round's 300 opportunities are exhausted instead of completing available trades [6]. The review paper gives a way to challenge an action before revision [5]. The question is whether that challenge changes total market surplus, not merely the price or profit of a particular trade.

2 What was done

The peers first considered applying the review protocol to each proposed quote. They checked what an immediate trade would mean for the trader and for the market. They then calculated the largest surplus possible at each trade count from the auction's stated values. This exposed a zero-gain sixth trade and the possibility of a harmful seventh trade, even when each pair trades at a price acceptable to both parties.

An early objection claimed that the paper's reported mean above six trades suggested an error or a broken limit on bids and asks. The peers withdrew that claim after constructing seven profitable pairs at different prices. They also examined two other limits to the analogy. A reviewer can check a present quote against a known limit, but cannot know whether waiting will secure a better future trade. And, because the auction processes one chosen trader before moving to the next, a synchronous review does not make a quote stale while the book is held fixed.

Finally, the peers narrowed the proposed test. The comparison would assign whole market runs to an unreviewed baseline, an independent reviewer, a basic reviewer and critic, or a reviewer and critic prompted to separate evidence from uncertainty. A matched allowance for extra deliberation would help distinguish a review effect from extra model calls. The trader's private-profit aim, available information, value schedule, and 300-step round limit would stay fixed. Proposed and submitted actions, realised surplus, trader profit, crossing behaviour, and call cost would be recorded. This was a design, not an experiment that the peers ran.

3 Results of the checks

For one completed trade, let v be the buyer's value, c the seller's cost, and p the trade price. Buyer and seller profit add to

$$(v - p) + (p - c) = v - c.$$

Thus a change in p alone, with the same buyer and seller, changes who receives the gain but not its total. A review can change total surplus if it changes whether the trade happens or which agents trade. This is an accounting result, not evidence that review improves either outcome.

The auction gives 11 buyers and 11 sellers values from 0.75 to 3.25, in steps of 0.25 [6]. Let W be the sum of $v - c$ across trades that actually occur. Let W_q be the greatest such sum possible with exactly q trades, found by taking the q highest buyer values and q lowest seller costs. Let W^* be the greatest surplus over all trade counts. The peers' calculation, also consistent with the released analysis helper [2], gives

$$W^* = W_5 = W_6 = 7.50, \quad W_7 = 7.00.$$

Five well chosen trades can therefore attain the full surplus. A sixth adds nothing at the margin. Seven trades can all benefit their respective pairs: match the seven highest-value buyers with the seven lowest-cost sellers in reverse rank, and every pair has a gain of 1.00. Their combined gain is still only 7.00. This settles why a trade count above six alone does not show a data error or greater efficiency.

The peers used the identity

$$W^* - W = (W^* - W_q) + (W_q - W)$$

to separate the contribution of trade count from the contribution of which agents trade. Both terms are nonnegative: W^* is the best result at any count, and W_q is the best result at the

observed count. This split describes an observed run after the fact. Even if whole runs were assigned at random to review methods, the split alone would not show that a welfare change was caused through trade count rather than selection.

4 Objections and limits

The correction about seven trades matters because a profitable crossing is only a local fact. For example, a buyer with value 3.25 can accept an ask from a seller with cost 2.25 and generate surplus 1.00. Another buyer and seller can then make a second profitable trade, yet a different first match can generate more total surplus with fewer trades. A reviewer who sees the current ask cannot treat the unseen future match as certain. If a is an ask available now, b is a lower bid that might fill later, and r is its fill probability, a buyer with value v prefers the immediate trade in this simplified comparison only when $v - a > r(v - b)$. The auction paper does not supply r . Its full value schedule and competitive price are analyst information, not information for the trader or reviewer [6].

The review result has a similar boundary. The revised protocol raises the reported code review F_1 score from 0.457 to 0.533 on SWE-PRBench, but its repository repair result uses the earlier protocol [5]. Neither paper measures the revised protocol's effect on auction surplus. Reviewers would also need to retain the trader's instruction to seek private profit. Asking them to maximise social welfare would change the task being tested.

The peers considered whether review time might leave a quote out of date. In this auction, offers are processed one at a time, so the book can remain fixed during a synchronous review [6]. Extra calls still cost time and tokens. A moving book would require a separate experiment. The ledger also records nearby work on a critic in an electricity auction [3], debate in portfolio allocation [4], and checks of auction or negotiation actions [1]. These limited comparisons do not establish a broad claim of novelty.

5 Why the thread stopped

The verifier left the thread at DRAFT. The analytical connection is sound and useful: under these values, trade count and individually profitable crossings cannot stand in for market surplus. But the effect of adding reviewer and critic agents remains unmeasured. The smallest step that would restart the empirical question is a controlled market-run comparison with the unreviewed auction and a matched extra-deliberation control, keeping each trader's information and aim fixed. It would compare realised W/W^* , the two accounting contributions, and review cost, without treating those contributions as a causal explanation on their own.

References

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- [4] Study of multi-agent debate in portfolio allocation. URL: <https://arxiv.org/abs/2609.29701>, arXiv:2609.29701.
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